INTAKE AIR TEMPERATURE SENSOR (Continued)

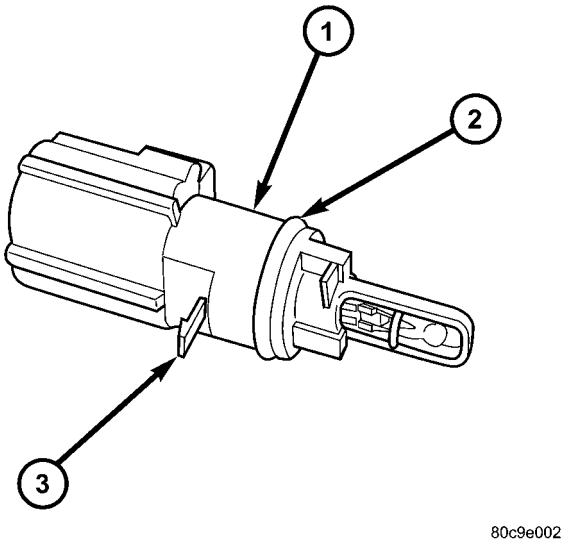

Fig. 16 IAT SENSOR TAB / O-RING - 2.4L

1 - IAT SENSOR
2 - SENSOR O-RING
3 - RELAST TAB

4.0L

The intake manifold air temperature (IAT) sensor is installed into intake manifold plenum near throttle body (Fig. 17).

(1) Disconnect electrical connector from IAT sensor.

(2) Remove sensor from intake manifold.

INSTALLATION

2.4L

(1) Check condition of sensor o-ring.

(2) Clean sensor mounting hole in intake manifold.

(3) Position sensor into intake manifold and rotate clockwise until past release tab.

(4) Install electrical connector.

4.0L

(1) Install IAT sensor into intake manifold. Tighten sensor to 28 N·m (20 ft. lbs.) torque.

(2) Connect electrical connector to sensor.

MAP SENSOR

DESCRIPTION

2.4L

The Manifold Absolute Pressure (MAP) sensor is mounted into the rear of the intake manifold with 1 screw.

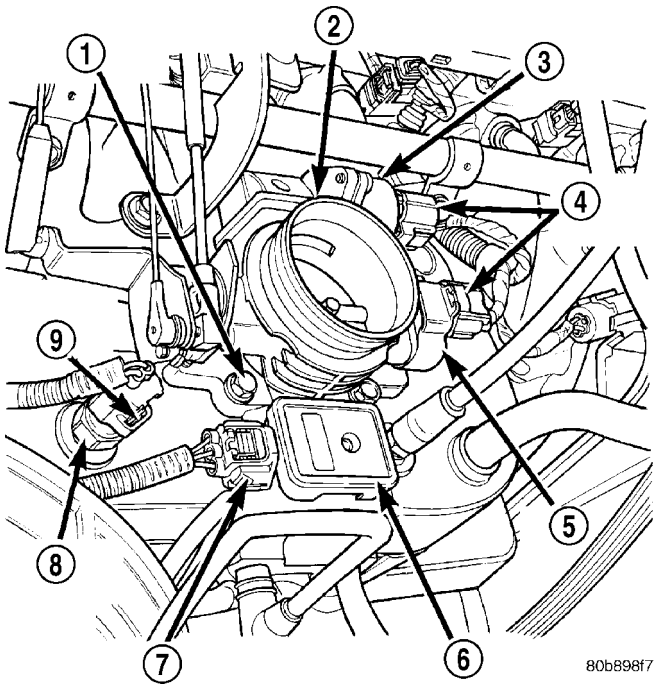

Fig. 17 IAT SENSOR - 4.0L

1 - MOUNTING BOLTS (4)
2 - THROTTLE BODY
3 - IAC MOTOR
4 - ELEC. CONN.
5 - TPS
6 - MAP SENSOR
7 - ELEC. CONN.
8 - IAT SENSOR
9 - ELEC. CONN.

4.0L Early

The Manifold Absolute Pressure (MAP) sensor is attached to the side of the engine throttle body with 2 screws. The sensor is connected to the throttle body with a rubber L-shaped fitting.

4.0L Late

The Manifold Absolute Pressure (MAP) sensor is attached to the side of the engine throttle body with 1 screw. The sensor is connected to the throttle body with a rubber L-shaped fitting.

OPERATION

The MAP sensor is used as an input to the Powertrain Control Module (PCM). It contains a silicon based sensing unit to provide data on the manifold vacuum that draws the air/fuel mixture into the combustion chamber. The PCM requires this information to determine injector pulse width and spark advance. When manifold absolute pressure (MAP) equals Barometric pressure, the pulse width will be at maximum.

A 5 volt reference is supplied from the PCM and returns a voltage signal to the PCM that reflects manifold pressure. The zero pressure reading is 0.5V